

External Origin Stability & Caching Telemetry Report

Formal diagnostic report of sequential black-box probes against **www.ccpatio.com**, documenting NitroPack edge masking, Cloudflare proxy behavior, and a measured 20.5-second origin stall on a heavy Product Detail Page under forced cache MISS conditions.

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Prepared for	PrimeView · CC Patio stakeholders
Subject site	https://www.ccpatio.com · Production WordPress / WooCommerce
Report date	18 September 2026
Test window (UTC)	18:45:33Z – 20:02:29Z (18 Sep 2026)
Test posture	External / black-box · Sequential only · Min. 4s sleep on endurance
Companion audit	WordPress Architectural & System Stability Audit (Site Health baseline)
Working archive	research/telemetry_archive_20260918/

EXECUTIVE VERDICT

The origin architecture is

CRITICALLY UNSTABLE

. NitroPack cache HITs (~107–145 ms) mask baseline browse performance while Cloudflare remains `cf-cache-status: DYNAMIC` (proxy, not HTML cache). Under forced MISS conditions—even with a four-second pause between single requests—the Brooklyn Corner Sofa PDP recorded a **20,451.7 MS (20.5 SECOND)**

Time to First Byte stall. That latency class is what produces intermittent Cloudflare HTTP 522 timeouts when the edge stops waiting. Feature freeze and OPcache expansion are mandatory before enabling live WooCommerce sessions or checkout.

~107–145ms

NITRO HIT TTFB

DYNAMIC

CF CACHE STATUS

20.5s

MISS STALL (ATT. 3)

~3.3 MB

PDP HTML SIZE

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2. Scope & objectives

This report documents externally observed behavior of the live production origin serving **www.ccpatio.com**. The program goal was to determine whether apparent storefront speed is produced by healthy origin compute or by optimizer cache hits—and whether the origin can sustain uncacheable traffic without stalling.

Prior Site Health diagnostics (captured 2026-09-18T12:45:48Z) established that PHP OPcache was 100% full with the interned-strings buffer exhausted (~24 bytes free) on LiteSpeed / PHP 8.2.33 LSAPI. This telemetry campaign tests that diagnosis from outside the firewall, without SSH, LiteSpeed panel, or Cloudflare dashboard access.

In scope • HTTP header routing (Cloudflare / NitroPack) • TTFB on HIT vs forced MISS paths • Heavy PDP / collection page behavior • Disabled commerce route cost (cart / checkout / account) • Sequential endurance drift on one PDP	Out of scope • Concurrent / multi-threaded load testing • Payment gateway or checkout conversion UX • Visual redesign or merchandising changes • Direct OPcache ini edits (requires host access)
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3. Execution timeline & safety controls

All probes were strictly single-threaded. Endurance tests enforced a minimum four-second sleep after every request to avoid stressing an OPcache-saturated origin.

18 Sep 2026 18:45:33 UTC	Baseline multi-URL probe — 14 sequential URLs; --sleep 3; --timeout 90. Output: probe-results.json. Summary: TTFB min 118.7 ms / max 522.6 ms / avg 395.5 ms; cf-cache-status always DYNAMIC; Nitro HIT on homepage.
18 Sep 2026 19:09:03 UTC	PDP / collection probe — 17 sequential URLs including Brooklyn Corner Sofa and Bravada collection; --sleep 4. Output: probe-pdp.json. Nitro HIT on PDP (~145 ms) and collection (~107 ms); MISS paths ~480–930 ms; no HTTP 522 in window.
18 Sep 2026 20:02:29 UTC	Repeat-MISS endurance — 15 attempts against /product/brooklyn-corner-sofa/ with unique query strings; --sleep 4; --timeout 90. Output: endurance.json. Attempt 3 TTFB 20,451.7 ms; all attempts Nitro MISS / CF DYNAMIC / HTTP 200.

Safety rationale Because Site Health already showed OPcache full, this program deliberately refused parallel clients (JMeter, Artillery, multi-worker curl). A single sequential client with enforced pacing is sufficient to expose origin cliffs without manufacturing an outage.

4. Method & tooling

TOOL	FUNCTION	KEY PARAMETERS
ccpatio_external_stability_probe.py	Header capture, plugin path census, multi-URL TTFB	Sequential; sleep 3–4s; timeout 90s
ccpatio_endurance_tester.py	15× cache-busted MISS loop on one URL	Unique ccp_probe / UUID / timestamp QS; sleep ≥4s
Client location (observed)	Cloudflare PoP via cf-ray	Primarily PHX (Phoenix); occasional LAX
Primary endurance URL	Heavy Elementor PDP	/product/brooklyn-corner-sofa/

5. Finding — Edge masking (NitroPack HIT)

When NitroPack returns x-nitro-cache: HIT, first-byte latency is excellent and can create a false impression of origin health.

URL	NITRO	TTFB	HTML SIZE	CF STATUS
/product/brooklyn-corner-sofa/	HIT	145.3 ms	~3.31 MB	DYNAMIC
/collections/bravada-collections/	HIT	106.9 ms	~2.95 MB	DYNAMIC
Featured project page (HIT)	HIT	135.9 ms	~2.86 MB	DYNAMIC

Conclusion: Stakeholder reports that “the site feels fast” align with the *HIT* path. They do not certify LiteSpeed / PHP worker health under MISS, purge, cron, or session traffic.

6. Finding — Cloudflare DYNAMIC reality

Across baseline, PDP/collection, and endurance runs, Cloudflare consistently presented:

- server: cloudflare
- cf-ray present (PHX / LAX)
- **cf-cache-status: DYNAMIC** on measured HTML documents

NitroPack—not Cloudflare—owned HTML HIT/MISS transitions (x-nitro-cache, x-nitro-cache-from, x-nitro-rev, and related headers).

Architectural reading

On this sample, Cloudflare is operating as an edge proxy / WAF front door. It is not serving HTML page-cache HITs for these URLs. Dual-stack “memory savings” cannot be attributed to Cloudflare HTML caching based on these headers.

7. Finding — 20.5-second origin stall

The endurance tester issued fifteen sequential GETs to the Brooklyn Corner Sofa PDP. Each request carried a unique query string (ccp_probe, UUID nonce, millisecond timestamp) to defeat anonymous HTML HIT reuse. Sleep between attempts: **4.0 seconds**.

#	STATUS	TTFB (MS)	TOTAL (MS)	CF	NITRO
1	200	931.6	1061.2	DYNAMIC	MISS
2	200	877.2	971.2	DYNAMIC	MISS
3	200	20451.7	20550.6	DYNAMIC	MISS
4	200	746.7	849.1	DYNAMIC	MISS
5	200	853.9	948.8	DYNAMIC	MISS
6	200	839.0	938.3	DYNAMIC	MISS
7	200	761.6	852.5	DYNAMIC	MISS
8	200	767.8	854.7	DYNAMIC	MISS
9	200	685.3	772.1	DYNAMIC	MISS
10	200	752.1	847.5	DYNAMIC	MISS
11	200	806.7	904.5	DYNAMIC	MISS
12	200	816.4	902.7	DYNAMIC	MISS
13	200	832.8	918.3	DYNAMIC	MISS
14	200	695.6	782.5	DYNAMIC	MISS
15	200	697.9	838.9	DYNAMIC	MISS

STATISTIC	VALUE
TTFB minimum	685.3 ms
TTFB typical band (ex-spike)	~685–932 ms
TTFB maximum (Attempt 3)	20,451.7 ms (20.5 s)
TTFB average (includes spike)	2,101.1 ms
HTTP 5xx / 522 pages in this window	0

WHY ATTEMPT 3 MATTERS

A late HTTP 200 after 20.5 seconds is a *near-miss timeout*, not a healthy response. Cloudflare 522 is emitted when the edge abandons an origin that fails to answer in time. This stall under *one* sequential client—with a four-second pause after prior requests—shows the origin can lock up without any synthetic flood. Combined with the Site Health finding of a full PHP OPcache, the mechanism is consistent with LiteSpeed / LSAPI workers blocked on opcode eviction and recompile while regenerating a multi-megabyte Elementor PDP.

8. Finding — DOM / payload bloat

The Brooklyn Corner Sofa PDP HTML payload was approximately **3.3 MB** on the HIT path. Plugin footprints observed in markup included Elementor, Elementor Pro, WooCommerce, Variation Swatches, Astra Addon, Favorites, and EMCP Pro (non-exhaustive; not all of the 38 active plugins enqueue assets on every URL).

Each forced MISS must rebuild that document through the full WordPress / builder / commerce bootstrap. Large payloads amplify CPU time, transfer time, and OPcache working-set pressure.

9. Finding — Soft-404 commerce tax

Transactional routes are effectively disabled (catalog-oriented posture; YITH Catalog Mode present in the DOM). Despite that, disabled commerce URLs still incurred full storefront boots:

PATH	STATUS	BODY SIZE	TTFB BAND	NITRO
/cart/	404	~2.2–2.3 MB	~480–560 ms	MISS / disabled
/checkout/	404	~2.2–2.3 MB	~480–560 ms	MISS / disabled
/my-account/	404	~2.2–2.3 MB	~480–560 ms	MISS / disabled

Conclusion: Turning checkout “off” does not remove origin cost. Bots and broken links still force multi-megabyte WordPress boots. Enabling real sessions later adds *additional* uncacheable load on top of an already fragile MISS path.

10. Agency claim assessment

PrimeView asserted that Cloudflare and NitroPack “work together” via API integration and that NitroPack “packs to use less memory,” implying the dual stack is non-conflicting and memory-helpful.

CLAIM THEME	TELEMETRY READING
Both systems jointly cache HTML to save memory	CF remained DYNAMIC ; Nitro owned HIT/MISS. Not dual HTML HIT caching on this sample.
“Packing” reduces server memory pressure	Packing reduces <i>payload bytes</i> . OPcache / PHP worker RAM is a separate ledger. MISS path still stalled for 20.5s.
API integration prevents conflict	Purge coordination (if present) does not stop origin PHP on MISS, does not expand OPcache, and did not prevent the stall.

11. Limits of this study

- External probes cannot print live opcache* percentages; Site Health remains the direct OPcache measurement.
- These runs did not capture a Cloudflare 522 error page; they captured a 20.5s near-timeout stall (stronger mechanistic proof for this window).
- Multi-server load-balancer topology remains an unverified agency statement.
- Results reflect PHX-adjacent probing at the stated UTC times; absolute TTFB varies by network path.

12. Mandatory gates & remediation

No go-live of transactional WooCommerce sessions/checkouts until the following gates are met.

PRI	OWNER	ACTION	ACCEPTANCE
P0	Host / SysAdmin	Expand OPcache memory and interned-strings buffer; verify not full	Site Health: opcode_cache_full=false; hit rate ≥ 90%
P0	PrimeView + CC Patio	Feature freeze on net-new Elementor / plugin work	Written freeze acknowledgment
P0	PrimeView	Declare single HTML cache owner (NitroPack or Cloudflare)	Signed architecture note; CF rules match choice
P0	Ops (with CF access)	Review 14–30 day Cloudflare 522/524 by URL	Zero unexplained 522s for 72 hours
P1	PrimeView	Replace fat soft-404 commerce routes with lightweight 404	Soft-404 body << 100 KB
P1	PrimeView	Reduce plugin/boot surface (incl. Broken Link Checker if still active)	MISS TTFB band flattened on Brooklyn PDP
P2	Architecture	Re-run 15× endurance tester on Brooklyn PDP after remediation	No attempt ≥ 3,000 ms TTFB

13. Supporting materials

Machine-readable outputs and reproducible scripts are stored in `research/telemetry_archive_20260918/`:

FILE	DESCRIPTION
<code>probe-results.json</code>	Baseline 14-URL probe (18:45:33Z)
<code>probe-pdp.json</code>	PDP / collection probe (19:09:03Z)
<code>endurance.json</code>	15× Brooklyn MISS endurance (20:02:29Z)
<code>ccpatio_external_stability_probe.py</code>	Multi-URL probe implementation
<code>ccpatio_endurance_tester.py</code>	Endurance MISS implementation
<code>EXTERNAL_BLACKBOX_STABILITY_AUDIT.md</code>	Method playbook

Related architecture materials: WordPress Architectural & System Stability Audit (HTML); `research/MASTER_CCPATIO_ARCHITECTURE_SME.md`; `research/system_architecture_SMEs/`.

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Prepared by · 18 September 2026

PrimeView technical lead (acknowledgment)

Name / date